Vehicle Parking Management System

Test Case Document

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Executed: 8 Nov 2026, commit 39e0e56 (feature/102-move-vehicle at 4ec9595, as merged to main in #104, plus testing-week tests). Jest 748 of 748 tests passed; browser tests 24 of 24 on Chrome and Edge locally; Firefox 12 of 12 via CI on main (run 41589654563). Result: 71 Pass, 1 Fail of 72 cases.

UT = Unit Test, IT = Integration Test, ST = System Test. Test Case ID format: <type>-<FR number>_<case number>.

FR-01 — Register an account and log in using valid credentials.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-01_01	User Authentication	Valid registration	Registration page is available	Enter valid name, email and password; click Register	Name=Arjun; Email=arjun@example.com; Password=Abc@123	Account should be created and a confirmation shown.	Account created; confirmation shown; driver can log in.	Pass
UT-01_02	User Authentication	Valid login	User account is registered	Enter registered email and password; click Login	Email=arjun@example.com; Password=Abc@123	User should be logged in and the home page shown.	Login succeeded and a session was opened; driver reached their dashboard.	Pass
IT-01_03	User Authentication	Login with wrong password	User account is registered	Enter registered email and a wrong password; click Login	Email=arjun@example.com; Password=wrong-pass	System should reject login and display an error.	Login refused with "Invalid email or password"; no session opened.	Pass
IT-01_04	User Authentication	Reject duplicate email	Email is already registered	Enter an already registered email; click Register	Email=ARJUN@example.com	System should display "email already in use".	Refused with "Email is already in use" (email match ignores case).	Pass
IT-01_05	User Authentication	Reject invalid email format	Registration page is available	Enter a malformed email; click Register	Email=abc@com	System should display an email validation error and not create the account.	abc@com refused with an email validation error; no account created.	Pass
ST-01_06	User Authentication	Lock account after 5 failed logins	User account is registered	Enter a wrong password 5 times in a row	Email=arjun@example.com; wrong password x5	Account should be temporarily locked (SEC-04).	5th wrong password locked the account with a message showing the time left.	Pass

FR-02 — Register, view, update and manage vehicle information.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-02_01	Vehicle Management	Add a new vehicle	User is logged in	Open My Vehicles; enter vehicle details; click Save	Reg No=KA01AB1234; Type=Car	Vehicle should be saved and listed.	Vehicle saved and listed. Type is chosen as Four-wheeler (car), since types match slot types.	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-02_02	Vehicle Management	View registered vehicles	User has registered vehicles	Open My Vehicles	N/A	List of the user's vehicles should be displayed.	Only the driver's own vehicles are listed.	Pass
IT-02_03	Vehicle Management	Reject duplicate vehicle	Vehicle is already registered	Enter the same registration number; click Save	Reg No=KA01AB1234	System should display "vehicle already exists".	Refused with "This vehicle already exists", even when typed with spaces or hyphens.	Pass
IT-02_04	Vehicle Management	Reject invalid registration number	User is logged in	Enter an invalid registration number; click Save	Reg No=1234	System should display a validation error.	1234 refused with a validation error.	Pass
IT-02_05	Vehicle Management	Update vehicle details	Vehicle exists	Select vehicle; change colour; click Save	Colour=Red to Blue	Updated details should be saved.	Colour changed from Red to Blue and saved.	Pass
ST-02_06	Vehicle Management	Updated vehicle persists after re-login	Vehicle exists	Update vehicle; log out; log in; open My Vehicles	Colour=Blue	Updated colour should be displayed.	Blue still shown after logging out and in again.	Pass

FR-03 — View the current availability and status of parking slots.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-03_01	Parking Availability	Display slot status	Slots exist in the system	Open Slot Availability	N/A	Slots should show Available, Occupied, Reserved or Unavailable.	Slots showed Available, Occupied, Reserved and Unavailable correctly.	Pass
UT-03_02	Parking Availability	Filter available slots	Slots with mixed statuses exist	Open Slot Availability; apply Available filter	Filter=Available	Only available slots should be listed.	Available filter returned only available slots; other filters also correct.	Pass
IT-03_03	Parking Availability	Availability after slot is occupied	A slot is assigned to a vehicle	Assign slot; refresh Slot Availability	Slot=P-12	Slot P-12 should show Occupied.	Slot showed Occupied immediately after entry.	Pass
IT-03_04	Parking Availability	Availability after slot is released	Vehicle has exited	Process exit; pay the fee; refresh Slot Availability	Slot=P-12	Slot P-12 should stay Occupied until the fee is paid, then show Available (UC-08).	Slot stays Occupied after exit until the fee is paid, then shows Available (UC-08).	Pass
IT-03_05	Parking Availability	Attendant views availability	Attendant is logged in	Open Slot Availability as attendant	N/A	Same real-time slot data should be displayed.	Attendant received the same live slot list as other roles.	Pass
ST-03_06	Parking Availability	Availability response time	Normal load conditions	Open Slot Availability; measure response time	N/A	Page should load within 2 seconds for 95% of requests (NFR-01).	95th percentile under 2 seconds over 20 requests with 300 slots (NFR-01).	Pass

FR-04 — Reserve an available parking slot for a specified period.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-04_01	Slot Reservation	Reserve an available slot	User is logged in; slot is available	Open slots; select slot; choose time period; click Confirm	Slot=P-05; 2 PM to 4 PM	Reservation should be created and the slot marked Reserved.	Reservation created; slot shows Reserved during the reserved time.	Pass
UT-04_02	Slot Reservation	Reservation requires login	User is logged out	Attempt to reserve a slot	Slot=P-05	User should be redirected to the login page.	API refused with 401; the page redirected to Log in (checked in browser).	Pass
IT-04_03	Slot Reservation	Reject already reserved slot	Slot is already reserved	Select the reserved slot; click Confirm	Slot=P-05 (reserved)	System should show an error and suggest another slot.	Refused as already reserved; page says "Choose another slot or time". No specific slot is named.	Pass
IT-04_04	Slot Reservation	Reject invalid time period	Slot is available	Select slot; enter end time before start time	Start=4 PM; End=2 PM	System should display a validation error.	End before start refused with a validation error.	Pass
IT-04_05	Slot Reservation	Reserved slot not given to another vehicle	Slot is reserved for 2 PM to 4 PM	Attempt entry into the reserved slot at 3 PM for a different vehicle	Slot=P-05; different vehicle	Assignment should be blocked (DR-08).	Other vehicle refused when P-100 was the only slot, and given P-101 when one was free (DR-08).	Pass
ST-04_06	Slot Reservation	Reservation shown in occupancy report	Reservation is created	Create reservation; open admin occupancy view	Slot=P-05	Occupancy count should include the reservation.	Occupancy report counted the reservation.	Pass

FR-05 — Record vehicle entry and create an active parking session.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-05_01	Vehicle Entry	Record vehicle entry	Attendant is logged in; slot is available	Enter vehicle number; click Confirm Entry	Reg No=KA01AB1234	Session should be created with the entry time recorded.	Active session created with the entry time recorded.	Pass
UT-05_02	Vehicle Entry	Reject unregistered vehicle	Vehicle is not registered	Enter unregistered vehicle number; click Confirm Entry	Reg No=KA99ZZ9999	System should display "vehicle not found".	Refused with "Vehicle not found".	Pass
IT-05_03	Vehicle Entry	Block second active session	Vehicle has an active session	Attempt entry for the same vehicle	Reg No=KA01AB1234 (active)	System should display "active session already exists" (DR-02).	Refused with "This vehicle already has an active parking session" (DR-02).	Pass
IT-05_04	Vehicle Entry	Block entry when no slot is free	No slots are available	Attempt vehicle entry	Reg No=KA01AB1234; no free slot	System should display "no available slot".	Refused with "No suitable parking slot is currently available".	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-05_05	Vehicle Entry	Entry timestamp is correct	Attendant is logged in	Process entry; check recorded entry time	Reg No=KA01AB1234	Entry time should match the system clock.	Entry time within 5 seconds of the database clock.	Pass
ST-05_06	Vehicle Entry	Entry completes within time limit	Normal conditions	Perform entry; measure duration	N/A	Entry should complete within 3 seconds (NFR-02).	Entry completed in well under 3 seconds (about 0.1 s) (NFR-02).	Pass

FR-06 — Assign an appropriate available slot and prevent duplicate assignments.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-06_01	Slot Assignment	Assign an available slot	Slot is available	Process entry; system assigns a slot	N/A	A suitable available slot should be assigned.	A free slot of the right type was assigned.	Pass
UT-06_02	Slot Assignment	Assign slot by vehicle type	Vehicle type is two-wheeler	Process entry for a two-wheeler	Type=Two-wheeler	A two-wheeler slot should be assigned (DR-07).	Two-wheeler was assigned a two-wheeler slot (DR-07).	Pass
IT-06_03	Slot Assignment	Occupied slot not reassigned	Slot is already occupied	Attempt to assign the occupied slot	Slot=P-08 (occupied)	Assignment should be rejected (DR-03).	Occupied slot was not assigned to another vehicle (DR-03).	Pass
IT-06_04	Slot Assignment	Unavailable slot not assigned	Slot is marked unavailable	Attempt to assign the unavailable slot	Slot=P-15 (unavailable)	Assignment should be rejected.	Disabled slot was not assigned.	Pass
IT-06_05	Slot Assignment	Attendant overrides slot	Several slots are available	Attendant selects a preferred slot; confirm	Slot=P-20	Selected slot should be assigned instead of the default.	Not implemented: entry always assigns the first free compatible slot; the attendant cannot choose one.	Fail
ST-06_06	Slot Assignment	Slot shows Occupied everywhere	Slot is assigned	Assign slot; open Slot Availability	Slot=P-08	Slot should show Occupied on every screen.	Occupied for driver, attendant and admin, in Track vehicle and in the report.	Pass

FR-07 — Identify the current parking location and status of a parked vehicle.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-07_01	Vehicle Tracking	Find a parked vehicle	Vehicle has an active session	Enter vehicle number; click Search	Reg No=KA01AB1234	Current slot and status should be displayed.	Slot and status shown to the attendant.	Pass
UT-07_02	Vehicle Tracking	Administrator tracks vehicle	Administrator is logged in	Search vehicle as administrator	Reg No=KA01AB1234	Vehicle location should be displayed.	Vehicle location shown to the administrator.	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
IT-07_03	Vehicle Tracking	Vehicle not parked	Vehicle is not parked	Enter vehicle number; click Search	Reg No=KA09ZZ0000	System should display "no active session found".	Shows "No active parking session found for this vehicle".	Pass
IT-07_04	Vehicle Tracking	Empty vehicle number	Attendant is logged in	Leave vehicle number blank; click Search	Reg No=blank	System should display a validation message.	Shows "Enter a registration number".	Pass
IT-07_05	Vehicle Tracking	Driver denied vehicle tracking	Driver is logged in	Attempt a staff-only action (track vehicle) as driver	Role=Driver	Access should be denied (SEC-02).	Driver refused with 403 (access denied).	Pass
ST-07_06	Vehicle Tracking	Tracking shows latest slot	Vehicle was moved to a new slot	Move the vehicle to another slot (Move vehicle); search vehicle	Reg No=KA01AB1234	The new slot should be shown, not the old one.	New slot shown, not the old one, after the attendant moved the vehicle with Move vehicle (#102).	Pass

FR-08 — Record vehicle exit and complete the parking session.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-08_01	Vehicle Exit	Record vehicle exit	Vehicle has an active session	Enter vehicle number; click Confirm Exit	Reg No=KA01AB1234	Exit time should be recorded; the session should wait for payment and complete once paid (UC-08).	Exit time recorded. Session waits for payment and completes once paid (UC-08).	Pass
UT-08_02	Vehicle Exit	Slot released after exit	Exit is processed	Process exit; pay the fee; check slot status	Slot=P-08	Slot status should change to Available once the fee is paid (SYS-12, UC-08).	Slot released to Available once the exit is paid, not at exit itself (UC-08, SYS-12).	Pass
IT-08_03	Vehicle Exit	Block exit without active session	Vehicle has no active session	Attempt exit for a vehicle that is not parked	Reg No=KA09ZZ0000	System should display "no active session" (DR-04).	Refused: no active parking session (DR-04).	Pass
IT-08_04	Vehicle Exit	Reject exit before entry time	Active session exists	Set an exit time earlier than the entry time	Entry=5 PM; Exit=3 PM	System should display a validation error (DR-09).	Exit earlier than entry refused with a validation error (DR-09).	Pass
IT-08_05	Vehicle Exit	Exit triggers fee calculation	Active session exists	Process exit; check the displayed fee	Entry=10:00; Exit=12:00	Fee should be calculated and shown at exit, before payment completes the session (UC-08, UC-09).	Fee calculated and shown at exit, before payment completes the session.	Pass
ST-08_06	Vehicle Exit	Completed session cannot be edited	Session is completed	Attempt to edit the completed session	Session ID=S-1001	Edit should be blocked (DR-10).	No route edits a session; a second exit is refused (DR-10).	Pass

FR-09 — Calculate parking duration and charges from the configured rates.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-09_01	Fee Calculation	Fee for whole hours	Entry and exit times are available	Process exit; system calculates fee	Entry=10:00; Exit=12:00; Rate=₹20/hr	Fee of ₹40 should be displayed.	Fee of ₹40.	Pass
UT-09_02	Fee Calculation	Fee for a partial hour	Entry and exit times are available	Process exit after a partial hour	Entry=10:00; Exit=10:45; Rate=₹20/hr	Fee should be rounded by the configured rule and displayed.	₹15 with a 15-minute billing step; ₹20 with a 60-minute step.	Pass
IT-09_03	Fee Calculation	Fee uses configured rate	Administrator has set the rate	Set rate; process exit	Rate=₹30/hr; Duration=1.5 hr	Fee of ₹45 should be displayed.	Fee of ₹45.	Pass
IT-09_04	Fee Calculation	Zero-duration stay	Entry and exit at the same time	Process exit immediately after entry	Entry=Exit time	Minimum fee should apply with no error.	Minimum fee charged; no error.	Pass
IT-09_05	Fee Calculation	Rate change applies to new exits	New rate is configured	Update rate; process a new exit	New rate=₹25/hr	Fee should use the updated rate.	New exit used the updated rate.	Pass
ST-09_06	Fee Calculation	Fee shown without delay	Normal load	Process exit; measure time to show fee	N/A	Fee should be shown within the 3-second exit flow (NFR-02).	Fee returned in well under 3 seconds (about 0.04 s) (NFR-02).	Pass

FR-10 — Make and record parking payments.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-10_01	Payment Management	Successful payment	Fee is generated	Select payment method; click Pay	Amount=₹40; Method=UPI	Payment should be recorded and the session marked Paid.	₹40 UPI payment recorded; session marked Paid.	Pass
UT-10_02	Payment Management	Payment options listed	Fee is generated	Open the payment screen	N/A	Available payment methods should be listed.	Cash, UPI and Card listed.	Pass
IT-10_03	Payment Management	Failed payment recorded	Payment gateway is available	Submit payment; gateway declines	Amount=₹40; invalid card	Status should be Failed and the user notified.	Decline recorded as Failed; user told; session stays unpaid. Simulated gateway.	Pass
IT-10_04	Payment Management	Reject expired card	Payment gateway is available	Enter an expired card; click Pay	Expired card number	System should display a decline error.	Expired card refused with an error; nothing recorded. Simulated gateway.	Pass
IT-10_05	Payment Management	Attendant takes payment	Attendant is logged in	Attendant selects payment method; confirms payment	Amount=₹40; Method=Cash	Payment should be recorded against the session.	₹40 cash recorded against the session by the attendant.	Pass
ST-10_06	Payment Management	Payment linked to session	Payment is completed	Complete payment; open the session record	Session ID=S-1001	Payment should be linked to the correct session (SYS-13).	Payment linked to the correct session, for that session's fee (SYS-13).	Pass

FR-11 — Maintain and display historical parking-session information.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-11_01	Parking History	View completed sessions	User has past sessions	Open Parking History	User account with history	Past sessions with fee and status should be listed.	Past sessions listed with fee and status, newest first.	Pass
UT-11_02	Parking History	Session record fields	Completed session exists	Open Parking History; open a session record	Session ID=S-1001	Vehicle, slot, entry/exit, duration, fee and status should be shown.	Vehicle, slot, entry, exit, duration, fee and status all shown.	Pass
IT-11_03	Parking History	Driver cannot edit history	Completed session exists	Attempt to edit a past session	Session ID=S-1001	Edit should be disabled or blocked (DR-10).	No route changes a session; the record stayed as it was (DR-10).	Pass
IT-11_04	Parking History	Empty history for new user	New user with no sessions	Open Parking History	New user account	A no-records message should be shown.	Empty list returned, no error; the page shows a no-records message.	Pass
IT-11_05	Parking History	Administrator views a user's history	Administrator is logged in	Open a selected user's history	User=Driver X	That user's history should be displayed.	Administrator opened the selected driver's history.	Pass
ST-11_06	Parking History	History with many sessions	User has 50+ sessions	Open Parking History; scroll or page through	User with 50+ sessions	All records should be reachable without errors.	Paging reached every one of the 50+ sessions once, no errors.	Pass

FR-12 — Manage users, vehicles, slots and rates, and generate parking reports.

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
UT-12_01	Administration and Reports	Generate occupancy report	Administrator is logged in	Open Reports; select Occupancy; click Generate	Date range=Today	Report should show current occupancy.	Report showed current occupancy for today.	Pass
UT-12_02	Administration and Reports	Add or remove a user	Administrator is logged in	Open User Management; add or remove a user	New user details	User list should update accordingly.	Admin added an attendant (who could log in) and removed a user; list updated.	Pass
IT-12_03	Administration and Reports	Only administrators see audit logs	Attendant is logged in	Attempt to open audit logs as attendant	Role=Attendant	Access should be denied (SEC-08).	Attendant refused with 403 (access denied).	Pass
IT-12_04	Administration and Reports	Rate change reflects in fee	Administrator is logged in	Update rate; process a new exit	New rate=₹25/hr	Fee should use the updated rate.	After the change to ₹25/hr, a 2-hour stay cost ₹50.	Pass
IT-12_05	Administration and Reports	Disable a parking slot	Administrator is logged in	Disable a slot; open Slot Availability	Slot=P-30	Slot should show Unavailable and not be assignable.	Slot showed Unavailable and was not assignable.	Pass

Test Case ID	Name of Module	Test case description	Pre-conditions	Test Steps	Test data	Expected Results	Actual Result	Test Result
ST-12_06	Administration and Reports	Audit log records admin actions	Administrator performs an action	Update a rate; open audit log	Action=rate update	Log should show the action, administrator and time (SEC-07).	Log showed the action, the administrator and the time.	Pass